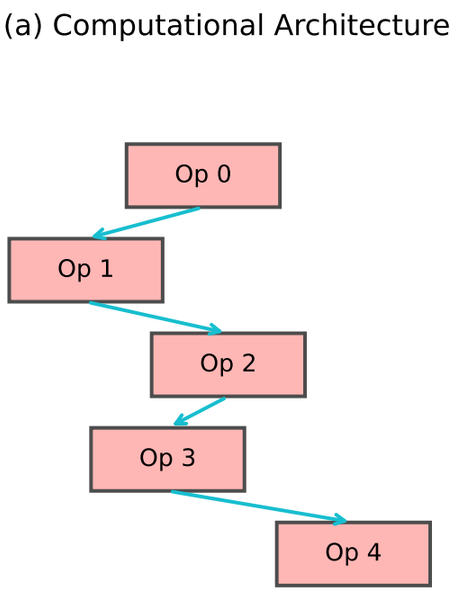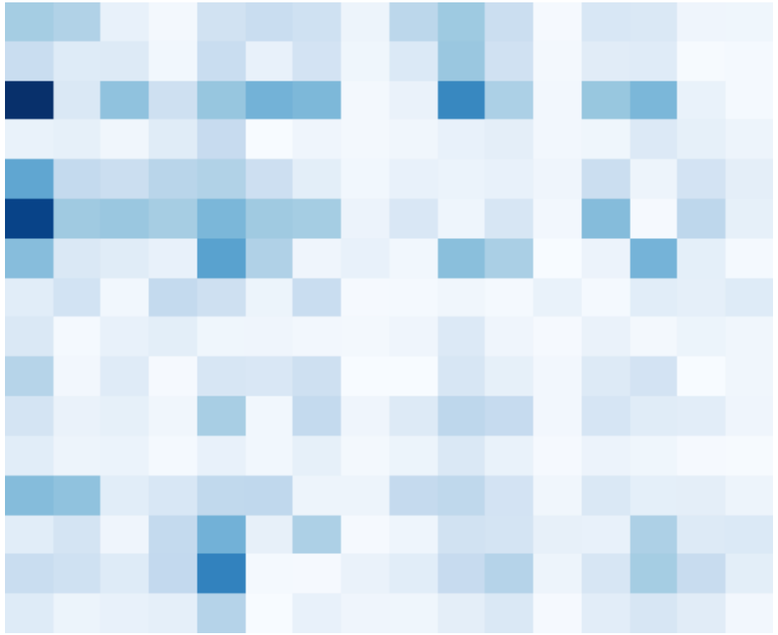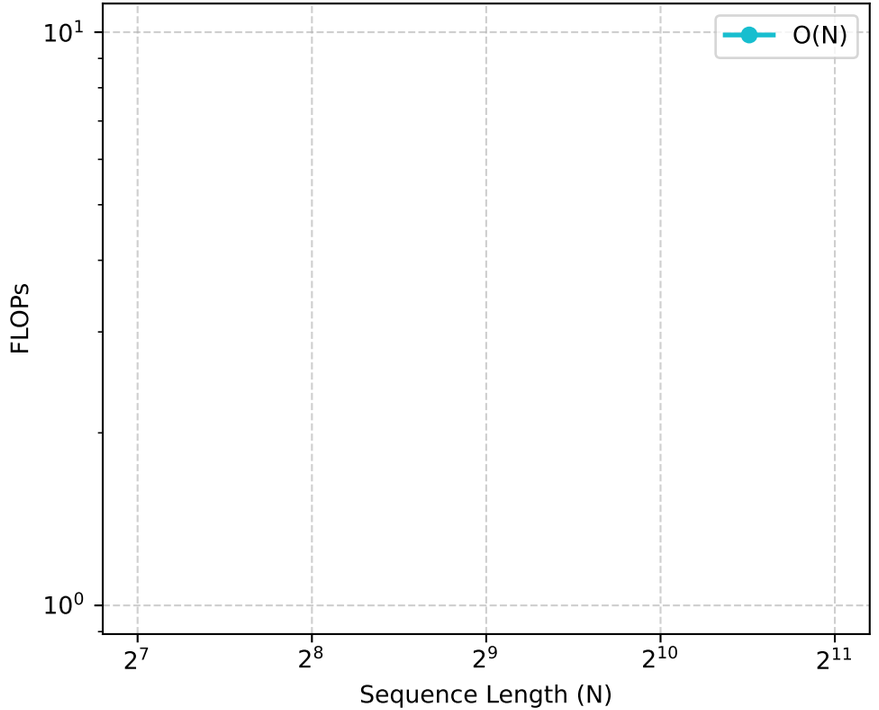Module Architecture: deserialization